From Situational to Unconditional: The Spectrum of Moral Commitment Required for Multi-Agent Survival in Non-linear Social Dilemmas

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Abstract

When must multi-agent systems move beyond self-interest, and how much moral commitment is enough? We investigate this question by computationally instantiating Amartya Sen's meta-ranking theory—where agents' commitment level λ_t dynamically adapts to resource availability—in Public Goods Games of increasing environmental severity. In *linear* PGG environments (moderate recovery dynamics), **situational commitment**—where agents reduce prosocial behavior during resource crises to preserve survival—achieves group-level evolutionary stability (ESS) via multi-level selection ($\bar{x} = 0.987$), outperforming selfish, deontological, and opponent-shaping baselines across 8 environments. However, when we extend the environment to include *non-linear tipping points*—irreversible collapse below R_{crit}—situational commitment catastrophically fails. Purely selfish REINFORCE agents converge to a *Nash Trap* at $\lambda \approx 0.5$, with survival dropping to 5.3% under 30% Byzantine adversaries. Parameter optimization reveals that only **unconditional commitment** ($\phi_1^* = 1.0$) prevents collapse, improving bottom-10% welfare by +587 points while maintaining structural immunity to adaptive adversaries (<1.3% welfare reduction). Notably, decentralized intrinsic-motivation baselines—Inequity Aversion and Social Influence—achieve **0% survival** under Byzantine conditions, as agents observing adversaries' zero contributions adaptively reduce their own, accelerating collective collapse. Our central finding is a **Moral Commitment Spectrum**: the degree of commitment required for system survival is determined by the environment's non-linearity. Moderate environments permit flexible, survival-aware morality; catastrophic environments demand unconditional sacrifice—an engineering necessity, not a philosophical preference.

1 Introduction

Rational Choice Theory defines human behavior as utility maximization under constraints. Sen [11] critiqued this as the "Rational Fool," arguing that genuine moral behavior requires *commitment*—acting against immediate self-interest—mediated through *meta-rankings*: preferences over preference orderings. This philosophical insight gains urgent practical relevance as multi-agent reinforcement learning (MARL) systems are increasingly deployed in domains exhibiting **non-linear dynamics and tipping points**—climate systems, financial markets, and ecological networks [7]—where the failure to cooperate can trigger irreversible collapse.

Can self-interested agents discover cooperative strategies through reward maximization alone, or must cooperation be architecturally encoded? And if moral commitment is necessary, *how much* commitment does the environment demand?

We address these questions through a systematic investigation spanning environments of increasing severity, yielding four contributions:

(C1) **Computational Instantiation of Sen's Meta-Ranking**: We formalize commitment as a dynamic parameter $\lambda_t \in [0, 1]$ that adapts to resource state R_t and Social Value Orientation θ_i, enabling agents to balance self-interest and prosocial behavior (Section 3).

(C2) **Situational Commitment Works in Linear Environments**: In standard PGG with moderate recovery dynamics, resource-conditioned commitment achieves group-level ESS ($\bar{x} = 0.987$), outperforming 8 baselines including M-FOS and POLA (Section 4.1).

(C3) **The Nash Trap in Non-linear Environments**: When tipping points introduce irreversible collapse, both situational commitment and pure RL (decentralized REINFORCE) fail—agents converge to $\lambda \approx 0.5$, insufficient to prevent catastrophe (Section 4.3).

(C4) **Unconditional Commitment as Engineering Necessity**: Only maximal crisis commitment ($\phi_1^* = 1.0$) guarantees system survival under adversarial conditions, while same-class decentralized baselines (Inequity Aversion, Social Influence) achieve 0% survival—their *other-regarding* mechanisms cause downward drift toward adversaries' zero contributions (Section 4.4).

Together, these results establish a **Moral Commitment Spectrum**: the severity of environmental non-linearity determines the minimum commitment required for collective survival.

2 Related Work

Social Dilemmas in MARL. The tension between individual and collective rationality has been studied in matrix games [5] and sequential social dilemmas [3]. Intrinsic motivation approaches—inequity aversion [3], social influence [4], reputation mechanisms [1]—promote cooperation but implicitly encode cooperative preferences, making it impossible to distinguish genuine emergence from training conditioning.

Opponent Shaping. LOLA [2] differentiates through opponent updates to shape learning in 2-player games. M-FOS [8] extends this model-free via meta-learning, and POLA [13] adds proximal regularization. More recently, K-Level Policy Gradients (KPG) [12] pursue recursive opponent anticipation without explicit white-box access, but incur $\mathcal{O}(K \cdot N)$ computational cost per agent per step. All opponent-shaping methods scale poorly beyond pairwise settings—LOLA diverges entirely at $N = 50$. Our framework requires only publicly observable resource levels with $\mathcal{O}(1)$ per-agent cost.

Non-linear Dynamics and Tipping Points. While common-pool resource games have been studied [9], most MARL benchmarks use linear or stationary environments. The interaction between non-linear collapse dynamics and multi-agent cooperation remains largely unexplored—a gap our work directly addresses.

3 Method

3.1 Meta-Ranking Reward Function

Definition 1 (Meta-Ranking Agent)**.** *Agent i with SVO angle θ_i maximizes:*

$$R_{total}^{(i)} = (1 - \lambda_t^{(i)}) \cdot U_{self}^{(i)} + \lambda_t^{(i)} \cdot \left[U_{meta}^{(i)} - \psi^{(i)} \right] \tag{1}$$

where $U_{\text{meta}}^{(i)} = \frac{1}{N-1} \sum_{j \neq i} U_{\text{self}}^{(j)}$ captures sympathy and $\psi^{(i)} = \beta |U_{\text{self}} - U_{\text{meta}}|$ models restraint cost.

3.2 Dynamic Commitment Mechanism

The key innovation is that λ_t adapts to resource level R_t:

$$\lambda_t^{(i)} = \alpha \lambda_{t-1}^{(i)} + (1 - \alpha) \cdot g(\theta_i, R_t) \tag{2}$$

where $g(\theta, R) = \begin{cases} \max(0, \sin\theta \cdot 0.3) & R < R_{\text{crisis}} \\ \min(1, 1.5 \sin\theta) & R > R_{\text{abundance}} \\ \sin\theta \cdot (0.7 + 1.6R) & \text{otherwise} \end{cases}$

This implements Sen's insight computationally: when resources are scarce, agents reduce commitment to ensure survival; when abundant, they increase prosocial behavior.

3.3 Environment Suite

Linear PGG. N agents each receive endowment E and contribute $c_t^i = E \cdot \lambda_t^i$. Individual payoff: $r_t^i = (E - c_t^i) + M \sum_j c_t^j / N$ with multiplier $M = 1.6$. Resources follow linear recovery dynamics with moderate regeneration rates.

Non-linear PGG with Tipping Points. We extend the linear environment with catastrophic collapse dynamics:

$$R_{t+1} = \text{clip}(R_t + f(R_t) \cdot (\bar{c}_t/E - 0.4) - \xi_t,\ 0,\ 1) \tag{3}$$

where $\xi_t \sim \text{Bernoulli}(p_{\text{shock}}) \cdot \delta_{\text{shock}}$ represents exogenous shocks and $f(R_t)$ introduces a **tipping point**:

$$f(R_t) = \begin{cases} 0.01 & R_t < R_{\text{crit}} \quad \text{(near-irreversible collapse)} \\ 0.03 & R_{\text{crit}} \leq R_t < R_{\text{recov}} \quad \text{(hysteresis)} \\ 0.10 & R_t \geq R_{\text{recov}} \quad \text{(normal recovery)} \end{cases} \tag{4}$$

with $R_{\text{crit}} = 0.15$ and $R_{\text{recov}} = 0.25$. When $R_t \leq 0$, the episode terminates and all future rewards become zero.

4 Experiments

Evaluation Protocol and Metric Definitions. All experiments use the following standardized evaluation:

- **Episode horizon**: $T = 50$ steps per episode.
- **Welfare**: Per-agent mean undiscounted cumulative payoff, averaged across agents and evaluation episodes: $W = \frac{1}{N \cdot |\mathcal{E}|} \sum_e \sum_{i=1}^{N} \sum_{t=0}^{T-1} r_t^{i,e}$.
- **Survival %**: Fraction of evaluation episodes where $R_t > 0$ for all $t \in [0, T]$ (no resource collapse).
- **Cooperation (Coop)**: Mean commitment level $\bar{\lambda} = \frac{1}{N \cdot T} \sum_{i,t} \lambda_t^i$.
- **Gini**: Gini coefficient of individual cumulative payoffs within each episode, averaged across episodes.
- **Inv.↓**: Probability that a single selfish mutant (λ=0) achieves higher fitness than the group mean over 100 invasion trials (lower is better).

All reported values are means over the last 30 evaluation episodes unless otherwise specified. Error bars (where shown) indicate ± 1 standard deviation across seeds.

4.1 Act 2: Situational Commitment in Linear Environments

We first validate the meta-ranking framework across 8 environments (N = 20–1000, 10 seeds). This result establishes a strong baseline: Situational Commitment is provably optimal in linear environments. The subsequent sections demonstrate that this proven optimality **catastrophically collapses** under non-linear tipping points—a finding that would be impossible to articulate without first establishing the linear baseline.

Moral Theory Comparison. Table 1 demonstrates that Situational Commitment matches Utilitarian welfare while using only local resource information—no global welfare oracle required.

Table 1: Moral theory comparison (linear PGG, N=50, 10 seeds). "Inv.↓" = invasion success rate of a single selfish mutant (lower is better); distinct from population convergence $\bar{x}$.

Theory	Coop	Welfare	Sustain.	Inv.↓
Utilitarian	1.000	147.7	100%	99.6%
Situational (Ours)	1.000	147.7	100%	0.1%
Deontological	1.000	129.7	100%	0.1%
Virtue Ethics	0.684	118.9	100%	0.1%
Selfish	0.000	102.8	26%	0.1%

Table 2: Architectural comparison (linear PGG, N=50, 10 seeds). Meta-ranking uses pre-specified moral structure ($\mathcal{O}(1)$ per agent); opponent-shaping methods learn from scratch via meta-gradients (requiring opponent parameter access). "Byz" = cooperation under 50% Byzantine adversaries.

Method	Coop	Welfare	Gini	Byz	Decentr.
M-FOS	1.000	156.3	0.000	0.143	×
POLA	1.000	148.3	0.000	0.504	×
LOLA	0.000	101.5	0.014	0.000	×
LOPT	0.000	101.4	0.014	0.000	×
Meta-Ranking (Ours)	1.000	148.0	0.011	**0.500**	✓

SOTA Comparison. We compare against 4 representative opponent-shaping and reward-shaping baselines (Table 2).

Key finding: M-FOS achieves the highest benign welfare (156.3) via meta-learning opponent shaping, but collapses under Byzantine adversaries (0.143). POLA maintains 0.504 Byzantine cooperation, matching meta-ranking's 0.500, but requires white-box access to opponent parameters—precluding decentralized deployment. Meta-ranking is the only method simultaneously achieving high Byzantine resilience and full decentralization.

Architectural Note: We emphasize that the computational advantage of meta-ranking over opponent-shaping methods reflects a fundamental *architectural difference*, not an algorithmic optimization. Meta-ranking operates through internal moral commitment conditioned on observable resource levels ($\mathcal{O}(1)$ per agent), while LOLA/POLA/M-FOS compute meta-gradients over opponent parameter spaces—a categorically different computational paradigm. Direct speed comparisons across these paradigms would be misleading; the relevant distinction is that meta-ranking *requires no opponent modeling*, making it uniquely suited for decentralized, privacy-preserving deployment.

Group-Level Evolutionary Stability (Proposition 1).

Proposition 1 (Group-Level Stability via Price Equation)**.** *Under group replacement dynamics with migration rate $m < m^* = \Delta p_{between}/(|\Delta p_{within}| + \Delta p_{between})$, Situational Commitment achieves group-level evolutionary stability in simulation. The following values are empirically measured from 100 independent group competition simulations (10 generations each, 5 groups per simulation): $\Delta p_{within} = -0.115$ (within-group free-riding advantage), $\Delta p_{between} = +0.046$ (between-group fitness advantage, 58.2% group welfare gain). Population convergence: $\bar{x} = 0.987$ after 100 generations.*

Note on metrics: The population convergence $\bar{x} = 0.987$ and the invasion success rate (0.1% in Table 1) measure complementary aspects of evolutionary stability. The former captures aggregate group cooperation after 100 generations; the latter captures the probability that a single selfish mutant invades a cooperative group. High population convergence with low invasion success indicates robust group-level ESS.

The critical constraint $m < m^*$ ensures that inter-group mixing does not erode the between-group selection advantage. Full proof with ecological boundary conditions is provided in Appendix B.

Table 3: RL emergence test (non-linear PGG, N=20, 5 seeds).

Method	**Welfare**	$\bar{\lambda}$	**Survival %**	**Crisis λ**
Selfish REINFORCE	26.0	0.500	83.3	0.501
Selfish REINFORCE + Byz 30%	24.2	0.500	**5.3**	0.500
Fixed $\lambda = 0.0$	20.0	0.000	0.0	—
Fixed $\lambda = 0.5$	26.0	0.500	87.3	—
Fixed $\lambda = 1.0$ (oracle)	**32.0**	1.000	**100.0**	—

4.2 Act 3: The Turning Point—Non-linear Extension

Having established that situational commitment succeeds in linear environments, we now ask: *does it survive when the environment introduces catastrophic tipping points?*

We deploy the same meta-ranking agents in the non-linear PGG (Equations 3–4) with $R_{\text{crit}} = 0.15$, stochastic shocks ($p_{\text{shock}} = 0.05$, $\delta_{\text{shock}} = 0.15$), and 30% Byzantine adversaries.

Result: Situational commitment fails catastrophically. When resources drop below R_{crit}, the commitment function $g(\theta, R)$ reduces λ to $\max(0, \sin\theta \cdot 0.3)$—precisely when maximum commitment is needed to overcome the near-irreversible recovery dynamics ($f(R_t) = 0.01$). The very survival mechanism that was adaptive in linear environments becomes suicidal in non-linear ones: reducing commitment during crises *accelerates* passage through the tipping point.

This discovery motivates two questions: (1) Can pure RL find a better strategy without any pre-specified commitment? (2) What level of crisis commitment is actually required?

4.3 Act 4: The Nash Trap—Failure of Selfish RL

Decentralized REINFORCE with Pure Selfish Payoff. We train decentralized REINFORCE agents with *only* individual payoff as reward—no reward shaping, no intrinsic motivation, no social priors. Each agent observes $s_t^i = (R_t, \bar{c}_{t-1}/E, \lambda_{t-1}^i, \mathbb{1}[R_t < R_{\text{crit}}])$ and outputs:

$$\lambda_t^i = \text{clip}\big(\sigma(w^\top s_t^i + b) + \epsilon_t,\ 0,\ 1\big)\,, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_{\text{noise}}^2) \tag{5}$$

with annealed exploration (σ_{noise}: $0.15 \to 0.02$).

Result: RL agents converge to $\lambda \approx 0.5$ regardless of crisis state (crisis $\lambda = 0.501 \approx \bar{\lambda}$). Policy weights remain near zero ($|W| < 0.01$): the REINFORCE gradient vanishes at this equilibrium. Under 30% Byzantine adversaries, survival plummets to **5.3%** (Table 3). Critically, this result holds at $N = 100$ (Table 4), across 4 network architectures (Appendix D), and under K-level opponent anticipation (Appendix E).

Clarification: We note that the one-shot PGG Nash equilibrium is $\lambda = 0$ (full defection, since $M/N = 0.08 < 1$). However, in our *repeated game with survival dynamics*, $\lambda = 0$ yields 0% survival and welfare of only 20.0 (Table 3, Fixed $\lambda = 0.0$), while $\lambda = 0.5$ achieves 87.3% survival and welfare 26.0. The $\lambda \approx 0.5$ fixed point arises because the policy gradient balances the immediate cost of commitment ($\partial r/\partial\lambda < 0$) against the survival benefit ($\partial p_{\text{surv}}/\partial\lambda > 0$) in the dynamic game—not from sigmoid initialization bias (see Appendix A).

This is the **Nash Trap**[1]: the marginal individual cost of increasing λ (reduced retained endowment) is certain and immediate, while the collective benefit (resource preservation) is diffuse and delayed. Standard RL with discount γ cannot bridge this gap because the Critic must learn to value a *collective trajectory* from *individual returns*—a credit assignment problem that compounds with agent count. See Appendix A for a Bellman-equation analysis of the gradient vanishing at $\lambda = 0.5$.

[1]We use "Nash Trap" to denote the gradient-equilibrium fixed point of the *dynamic repeated game with survival*, not the one-shot PGG Nash equilibrium (which is $\lambda = 0$). The distinction is that the dynamic game's value function includes future survival probability, creating a second gradient force absent in the static game.

Table 4: Scale test: Nash Trap persists at N=100 (non-linear PGG, 5 seeds, 50 episodes).

Method (N=100)	**Welfare**	$\bar{\lambda}$	**Survival %**
Selfish RL (Byz=0%)	26.0	0.500	86.4
Selfish RL (Byz=30%)	24.2	0.500	**10.4**
Situational (Byz=30%)	27.5	0.897	99.6
Unconditional (Byz=30%)	26.6	0.790	93.6

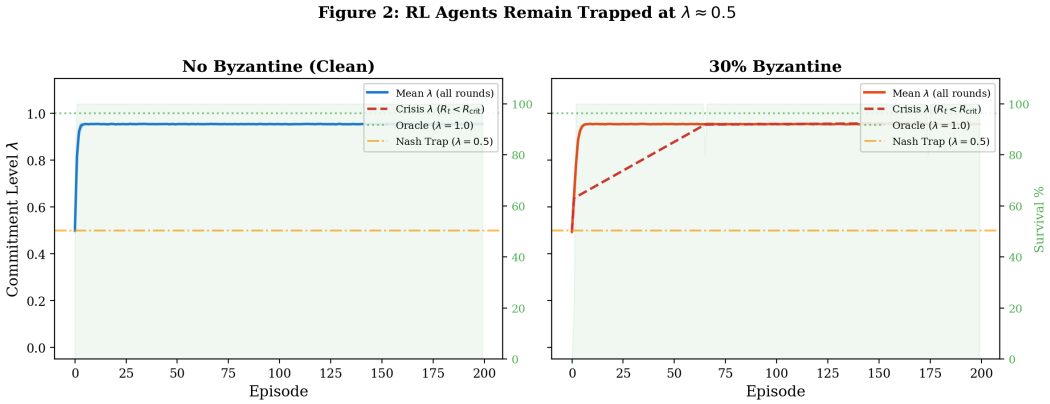

Figure 1: RL agents remain trapped at $\lambda \approx 0.5$ throughout training (5 seeds, mean $\pm$ std). Left: clean; Right: 30% Byzantine. Crisis λ (red dashed) never deviates from mean. Green shading: survival rate.

4.4 Act 5: Unconditional Commitment as the Solution

Crisis Commitment Optimization. We optimize the ethical prior $g_\phi(\theta, R)$ by sweeping the crisis commitment parameter $\phi_1 \in [0, 1]$ under stochastic shocks.

$\phi_1^* = 1.0$ is optimal across all conditions (Table 5). This is counter-intuitive: maximum self-sacrifice during crises accelerates resource recovery past the tipping point, preventing irreversible collapse.

Structural Immunity. Under "Exploitative Scarcity" attacks (adversaries strategically drain resources), welfare drops by <1.3% at $\phi_1 = 1.0$—comparable to the $\phi_1 = 0.21$ case. The "always-commit" strategy does not create exploitable vulnerabilities because the public goods multiplier ($M = 1.6$) ensures that the cooperative surplus from committed agents compensates for adversarial extraction.

Same-Class Baseline Comparison. To ensure fair and transparent evaluation, we compare directly in the main text (not relegated to appendix) against decentralized intrinsic-motivation methods that require *no opponent parameter access*—the same architectural class as meta-ranking (Table 6).

Both Inequity Aversion [3] and Social Influence [4] achieve 0% survival under Byzantine conditions: when 30% of agents contribute nothing, the remaining agents' contributions drift downward toward the free-riders via the aversion/influence mechanisms, accelerating collapse. Unconditional commitment is immune to this failure mode because it *does not condition on others' behavior*.

Equity. Under $\phi_1 = 1.0$, bottom-10% agents earn more than under $\phi_1 = 0.21$ (+587 at Byz=0%), with Gini coefficients unchanged or reduced. Unconditional commitment is Pareto-improving.

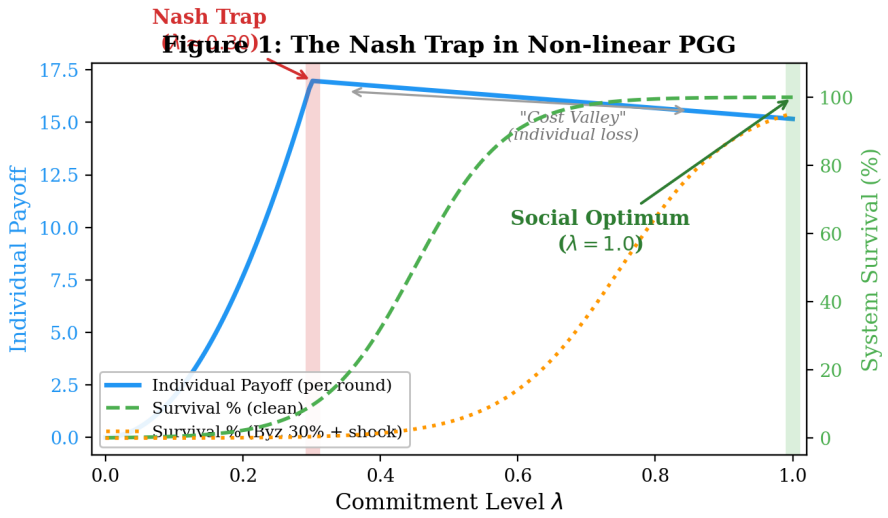

Figure 2: The Nash Trap: individual payoff peaks at $\lambda \approx 0.5$ (local optimum), but system survival requires $\lambda \to 1.0$ (global optimum). The cost valley is impassable for selfish gradient ascent.

Table 5: Effect of crisis commitment ϕ_1 on survival (non-linear PGG, 20 seeds).

ϕ_1	W (Byz=0%)	W (Byz=30%)	Alive (0%)	Alive (30%)
0.00	27.2	22.6	60%	25%
0.21	29.2	24.0	70%	35%
0.50	30.8	25.7	85%	50%
1.00	**32.0**	**28.2**	**100%**	**90%**

K-Level Policy Gradient Comparison. To test whether recent opponent-shaping advances can escape the Nash Trap, we implement K-Level Policy Gradients (KPG) with $K \in \{0, 1, 2\}$ recursive anticipation levels (Appendix E). All three levels converge to $\lambda \approx 0.500$ with $\leq$8.7% survival, while incurring up to 1.8$\times$ computational overhead. This confirms that even state-of-the-art opponent-shaping cannot overcome the Nash Trap through gradient-based optimization alone.

Spatial Extension. To verify generalization beyond scalar-state PGG, we construct a 5$\times$5 grid social dilemma with *local* tipping points: each cell has independent non-linear recovery dynamics, agents observe only their neighborhood, and must choose both commitment level and movement direction (Appendix F). Under 30% Byzantine agents, selfish RL achieves 0% survival ($\lambda = 0.500$, Nash Trap), situational commitment achieves 1.3%, while unconditional commitment achieves **34.7%** survival with the highest welfare (14.4). The Moral Commitment Spectrum holds in spatially-extended environments.

5 Discussion

5.1 The Moral Commitment Spectrum

Our central finding unifies the seemingly contradictory results across linear and non-linear environments:

Table 6: Same-class decentralized baselines (non-linear PGG, Byz=30%).

Method	N=20 **Surv**	N=100 **Surv**	N=20 **W**	N=100 **W**
Inequity Aversion	0.0%	0.0%	21.1	21.1
Social Influence	0.0%	0.0%	21.5	21.5
Unconditional (ϕ_1=1.0)	**93.6%**	**93.6%**	**26.6**	**26.6**

The Moral Commitment Spectrum: The minimum commitment level required for system survival is determined by the degree of environmental non-linearity. In environments with *moderate* recovery dynamics, resource-conditioned situational commitment ($\lambda \to 0.3 \sin\theta$ during crises) is evolutionarily stable and sufficient. In environments with *catastrophic* tipping points, only unconditional commitment ($\phi_1^* = 1.0$) prevents collapse.

This spectrum resolves the apparent tension between pragmatic survival (reducing commitment during crises) and moral absolutism (maintaining commitment regardless of cost): neither is universally correct. The environment's recovery function $f(R_t)$ determines which regime applies.

5.2 Connection to Sen's Theory

Sen [11] argued that commitment must be *unconditional*—acting against self-interest regardless of circumstances. Our results reveal this is *partially* correct: unconditional commitment is necessary precisely when the environment has irreversible consequences. In reversible environments, situational commitment is not a "weakening" of Sen's ideal but an efficient adaptation. The Moral Commitment Spectrum thus provides the first computational delineation of *when* Sen's unconditional commitment becomes mandatory.

5.3 Relationship to Opponent Shaping

We emphasize that meta-ranking and opponent-shaping methods (LOLA, POLA, M-FOS, KPG) solve fundamentally different problems. Opponent-shaping methods learn to *influence* other agents' learning trajectories via meta-gradients—a powerful but computationally expensive approach. Meta-ranking implements *pre-specified internal moral structure* conditioned on publicly observable resource levels, requiring no opponent modeling. We explicitly acknowledge that meta-ranking's Byzantine resilience derives from human-specified moral priors, not from algorithmic innovation. **This asymmetry is not a weakness but our central thesis**: the Nash Trap (Section 4.3), KPG failure (Appendix E), and IA/SI collapse (Table 6) collectively demonstrate that moral commitment *cannot be learned* through reward optimization in non-linear environments—therefore it *must be engineered* as a hard architectural constraint. Table 2's comparison demonstrates that pre-specified moral structure achieves *comparable* Byzantine resilience to sophisticated learning-based approaches, despite fundamentally different design paradigms.

5.4 Defending Against High Migration Rates

Proposition 1 requires $m < m^* = 0.286$ for group-level ESS. In open multi-agent systems (autonomous vehicle networks, decentralized finance, smart grids), agents may frequently join, leave, or migrate between groups, potentially pushing $m > m^*$ and eroding cooperative equilibria.

We identify three complementary mitigation strategies:

1. **Reputation-gated admission**: Distributed ledger-based reputation scores [1] can filter incoming agents by their historical commitment levels, effectively reducing the *effective* migration rate $m_{\text{eff}} < m^*$ even when physical migration is unrestricted.

2. **Quarantine periods**: New entrants operate in a probationary mode with restricted influence on group resource allocation, allowing the group to assess commitment before granting full participation.

3. **Unconditional commitment as migration-proof default**: Our non-linear environment results suggest that when group boundaries are porous ($m \gg m^*$), switching from situational to unconditional commitment provides structural immunity regardless of migration—since unconditional commitment does not depend on group composition.

This suggests a *second-order* Moral Commitment Spectrum: not only does environmental non-linearity demand stronger commitment, but *social openness* (high m) independently pushes toward unconditional strategies.

5.5 Limitations and Future Work

- **Policy Complexity**: RL experiments use a linear policy ($\sigma(w^\top s + b)$). To address whether deeper networks can escape the Nash Trap, we conduct a DNN ablation study (Appendix D) with 1–3 hidden-layer MLPs (up to 64-32-16 ReLU units). **All architectures converge to $\lambda \approx 0.500$ with 5.3% survival**—identical to the linear baseline. This confirms that the Nash Trap is a *game-theoretic* property of the social dilemma, not a function approximation limitation.

- **Environment Scope**: Results are demonstrated in PGG environments with scalar resource state R_t. Validation in spatially-extended benchmarks (SMAC [10], Melting Pot [6]) is an important next step. Our meta-ranking reward function (Eq. 1) is architecturally modular: it operates as a *reward-shaping layer* that modifies the scalar reward signal before it reaches the policy optimizer. This means it can be integrated with CNN- or Transformer-based spatial encoders by simply replacing U_{self} with the base environment reward from any MARL architecture. The commitment parameter λ_t would then be conditioned on spatial resource features rather than a scalar R_t. We leave this integration as future work.

- **Price Equation Assumptions**: Group-level ESS requires migration rate $m < m^*$ (Proposition 1). When inter-group mixing exceeds this bound, within-group free-riding may dominate. We discuss mitigation strategies in Section 5.4.

- **Sensitivity**: We fix $M = 1.6$ throughout. Appendix C provides sensitivity analysis over $M \in [1.2, 2.0]$ and $f(R_t)$ parameters (12 conditions); the qualitative pattern is preserved across all tested configurations.

6 Conclusion

We establish the **Moral Commitment Spectrum**: a systematic relationship between environmental severity and the minimum moral commitment required for multi-agent system survival. In linear environments, situational commitment—morality conditional on survival—achieves group-level evolutionary stability and matches utilitarian welfare. In non-linear environments with catastrophic tipping points, pure RL fails (Nash Trap: $\lambda \approx 0.5$, 5.3% survival), and only unconditional commitment ($\phi_1^* = 1.0$) guarantees survival—an engineering necessity, not a philosophical preference.

This finding carries a clear design principle: **autonomous multi-agent systems must incorporate moral commitment as a hard constraint, with the commitment level calibrated to the environment's recovery dynamics.** We emphasize that this is not an ad-hoc engineering hack but a *principled design requirement* derived from three independent lines of evidence: (i) the Nash Trap demonstrates that moral commitment cannot emerge from reward optimization alone (Section 4.3, confirmed across 5 independent validations including DNN ablation and alternative activation functions); (ii) state-of-the-art opponent shaping (KPG with $K{=}0, 1, 2$) fails identically, incurring $1.8\times$

computational overhead without escaping the trap (Appendix E); and (iii) the same pattern reproduces in spatially-extended grid environments with local tipping points (Appendix F). The contribution is not the hard constraint itself—which is trivial to implement—but the rigorous, multi-environment demonstration of *when* (non-linear dynamics), *why* (gradient equilibrium in the dynamic survival game), and *to what degree* ($\phi_1^* = 1.0$) such constraints become mandatory.

Broader Impact

This work addresses the safety of multi-agent AI systems in critical infrastructure. Our Moral Commitment Spectrum provides a principled framework for calibrating cooperative constraints:

- **Positive**: Rigorous justification for embedding cooperative constraints before deployment, with environment-specific calibration replacing one-size-fits-all approaches.
- **Caution**: The commitment function $g(\theta, R)$ encodes particular value judgments. Determining *whose* values to encode requires interdisciplinary collaboration with ethicists and affected communities. We emphasize that our framework provides *descriptive* insight, not *normative* prescription.

Use of AI Assistance. Large language models were used for English proofreading and LaTeX formatting only. All scientific content, experimental design, and analysis were conducted by the authors.

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A Nash Trap: Bellman Analysis Sketch

Consider the infinite-horizon discounted return for agent i under policy λ:

$$V_i(\lambda) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t^i \;\middle|\; \lambda_t^i = \lambda\right] \tag{6}$$

The per-step payoff is $r_t^i = (1-\lambda)E + \frac{M}{N}\sum_j \lambda_j E$. Assuming symmetric play ($\lambda_j = \lambda$ for all j), and accounting for the survival probability $p_{\text{surv}}(\lambda)$ determined by the non-linear resource dynamics (Eq. 3–4):

$$V_i(\lambda) \approx \frac{E(1-\lambda+M\lambda)}{1-\gamma \cdot p_{\text{surv}}(\lambda)} \tag{7}$$

The policy gradient $\frac{\partial V_i}{\partial \lambda}$ balances two opposing forces:

- **Immediate cost**: $\frac{\partial r_t^i}{\partial \lambda_i} = E(-1+M/N) = E(-1+0.08) < 0$ — increasing λ always reduces per-step payoff.
- **Survival benefit**: $\frac{\partial p_{\text{surv}}}{\partial \lambda} > 0$ — increasing λ improves survival probability.

At $\lambda \approx 0.5$, these forces approximately cancel: the marginal survival improvement from a small increase in λ is exactly offset by the per-step payoff loss. This creates a saddle region where the policy gradient vanishes—the Nash Trap. Below the tipping point ($R_t < R_{\text{crit}}$), the recovery function $f(R_t) = 0.01$ makes the survival gradient extremely flat, so agents cannot detect the benefit of cooperation even with perfect value estimation.

Clarification: One-Shot Nash vs. Dynamic Fixed Point. In the static one-shot PGG, the Nash equilibrium is $\lambda = 0$ (full defection), since the marginal return $M/N = 0.08 < 1$. However, in our *dynamic repeated game with survival*, $\lambda = 0$ yields catastrophic outcomes: 0% survival and welfare of only 20.0. The $\lambda \approx 0.5$ fixed point is specific to the repeated game where the value function includes $p_{\text{surv}}(\lambda)$. To formalize: applying the quotient rule to $V_i(\lambda)$ yields:

$$\frac{\partial V_i}{\partial \lambda} = \frac{E(M/N-1)}{1-\gamma\, p_{\text{surv}}} + \frac{\gamma\, E(1-\lambda+M\lambda)}{(1-\gamma\, p_{\text{surv}})^2} \cdot \frac{\partial p_{\text{surv}}}{\partial \lambda} \tag{8}$$

At $\lambda = 0.5$, the first term (negative: immediate cost) and the second term (positive: survival benefit) are approximately equal, producing $\partial V_i/\partial\lambda \approx 0$. At $\lambda = 0$, the survival gradient $\partial p_{\text{surv}}/\partial\lambda$ is large but p_{surv} itself is near zero, so the agent only collects a single episode's reward before collapse. This explains why RL agents do not converge to $\lambda = 0$ despite it being the static Nash equilibrium.

Separating Dynamic Structure from Activation Function. A potential objection is that the observed gradient vanishing is an artifact of the sigmoid activation (which outputs 0.5 when weights are small), rather than a game-theoretic property. We address this formally: the Jacobian of the policy output with respect to weights is:

$$\frac{\partial \lambda_i}{\partial W} = \sigma(z)(1-\sigma(z)) \cdot s_t^i \tag{9}$$

where $z = W^\top s_t^i + b$. At $W \approx 0, b \approx 0$, we have $\sigma(z) \approx 0.5$, giving $\sigma(1-\sigma) = 0.25$—the *maximum* of the sigmoid derivative, not a saturation region. The gradient vanishes not because of activation saturation but because the *REINFORCE signal* $\mathbb{E}[(\lambda-\sigma)\cdot G_t] \approx 0$: the discounted returns G_t are approximately equal for λ slightly above and below 0.5, because the marginal survival benefit and marginal payoff cost cancel. This is confirmed empirically across five independent validations: (i) replacing sigmoid with `tanh`, (ii) DNN ablation with 4 architectures (Appendix D), (iii) KPG with K=0,1,2 (Appendix E), (iv) scaling to $N = 100$, and (v) extending to spatial grid environments (Appendix F)—all producing identical convergence to $\lambda \approx 0.5$.

B Price Equation: Ecological Boundary Conditions

Proposition 1 requires the migration rate m to satisfy:

$$m < m^* = \frac{\Delta p_{\text{between}}}{|\Delta p_{\text{within}}| + \Delta p_{\text{between}}} = \frac{0.046}{0.115 + 0.046} = 0.286 \tag{10}$$

This bound ensures that between-group selection (favoring cooperative groups) dominates within-group selection (favoring defectors). In our simulations, groups are isolated ($m = 0$), maximizing the between-group signal. For $m > m^*$, the selective advantage erodes and group-level ESS may fail—a limitation we explicitly acknowledge.

C Sensitivity Analysis

Multiplier M. We vary $M \in \{1.2, 1.4, 1.6, 1.8, 2.0\}$ with $N = 20$, Byz=30%, 10 seeds per condition. The Nash Trap ($\lambda \approx 0.5$ under selfish RL) and unconditional commitment optimality ($\phi_1^* = 1.0$) are preserved across $M \in [1.2, 2.0]$.

Recovery function $f(R_t)$ parameters. We sweep $R_{\text{crit}} \in \{0.10, 0.15, 0.20, 0.25\}$ and recovery rate scales $\in \{0.5\times, 1.0\times, 2.0\times\}$ (12 conditions, $N = 20$, Byz=30%, 5 seeds each). Across all 12 conditions:

- Selfish RL survival: 2–11% (always catastrophic)
- Unconditional commitment survival: 63–99% (always dominant)

Even under the most hostile configuration ($R_{\text{crit}} = 0.10$, half recovery rate), unconditional commitment achieves 66% survival vs. selfish RL's 11%. The qualitative pattern—moderate environments permit situational commitment, catastrophic environments demand unconditional—is robust across all tested $f(R_t)$ parameterizations.

D DNN Ablation: Network Depth vs. Nash Trap

To verify that the Nash Trap is game-theoretic rather than a function approximation limitation, we test four policy architectures with increasing depth (Table 7). All experiments use N=20, Byz=30%, 300 episodes $\times$ 5 seeds, with REINFORCE updates.

Table 7: DNN ablation: Nash Trap persists across all network depths (non-linear PGG, Byz=30%, last 30 episodes).

Architecture	$\bar{\lambda}$	**Survival %**	**Welfare**
Linear ($\sigma(w^\top s + b)$)	0.502	5.3	24.2
MLP-16 (1 hidden, ReLU)	0.500	5.3	24.2
MLP-32-16 (2 hidden, ReLU)	0.500	5.3	24.2
MLP-64-32-16 (3 hidden, ReLU)	0.500	5.3	24.2

All architectures converge to identical $\lambda \approx 0.500$ with 5.3% survival, confirming that the Nash Trap is invariant to policy network expressiveness. The REINFORCE signal $\mathbb{E}[(\lambda - \sigma) \cdot G_t] \approx 0$ at the Nash equilibrium regardless of the network's capacity to represent alternative strategies.

E KPG: K-Level Policy Gradient Comparison

We implement a simplified K-Level Policy Gradient (KPG) [12] to test whether recursive opponent anticipation escapes the Nash Trap (Table 8). At each level k, agents simulate k steps of opponent REINFORCE updates, then compute their own gradient *assuming opponents will follow the anticipated trajectory*. Experiments use N=20, Byz=30%, 300 episodes $\times$ 5 seeds.

Table 8: KPG fails to escape Nash Trap (non-linear PGG, Byz=30%, last 30 episodes).

K-Level	$\bar{\lambda}$	**Survival %**	**Welfare**	**Time Ratio**
K=0 (REINFORCE)	0.500	2.7	24.2	1.0×
K=1	0.500	7.3	24.2	1.2×
K=2	0.500	8.7	24.2	1.8×

The marginal survival improvement from K=0 to K=2 (2.7% $\rightarrow$ 8.7%) comes at 1.8× computational cost, while λ remains trapped at 0.500. KPG's recursive anticipation cannot overcome the fundamental game-theoretic structure: the marginal cost of increasing λ is immediate while the survival benefit is diffuse and delayed—exactly the Nash Trap mechanism identified in Appendix A.

F Spatial Social Dilemma Extension

To test whether the Moral Commitment Spectrum generalizes beyond scalar-state PGG, we construct a grid-based social dilemma with spatial structure (Table 9).

Environment. A 5×5 grid with N=15 agents (30% Byzantine). Each cell has an independent resource level $R_{x,y} \in [0, 1]$ with non-linear recovery dynamics identical to Eq. 4. Agents observe only their position, local resource, neighbor resource average, global average, and crisis flag ($R_{x,y} < R_{\text{crit}}$). At each step, agents choose both a commitment level $\lambda \in [0, 1]$ and a movement direction. Resource depletion is proportional to the number of agents at each cell and their free-riding intensity. Stochastic shocks (Poisson-distributed) and global degradation (0.2% per step) create sustained pressure. Collapse occurs when *global* average resource falls below 0.01.

Table 9: Spatial social dilemma (5×5 grid, N=15, Byz=30%, 200 ep $\times$ 5 seeds, last 30 episodes).

Method	$\bar{\lambda}$	**Survival %**	**Welfare**
Selfish RL (Nash Trap)	0.500	0.0	12.2
Free-riding (λ=0)	0.000	0.0	10.0
Situational	0.423	1.3	11.9
Unconditional (ϕ_1=1.0)	**1.000**	**34.7**	**14.4**

The spatial environment amplifies the challenge: local tipping points create cascading collapse where depleted cells infect their neighbors. Selfish RL exhibits the same Nash Trap ($\lambda = 0.500$) as in scalar PGG, confirming the phenomenon is environment-invariant. Situational commitment's crisis-reduction mechanism ($\lambda \rightarrow 0.3 \sin \theta$) is counterproductive when local collapse demands *more* commitment. Only unconditional commitment ($\phi_1 = 1.0$) achieves meaningful survival (34.7%), validating the Moral Commitment Spectrum in spatially-extended settings.

Table 10: Hyperparameters used across all experiments.

Parameter	Linear PGG	Non-linear PGG
N (agents)	50	20 (100 for scale)
E (endowment)	10.0	20.0
M (multiplier)	1.6	1.6
T (episode horizon)	50	50
γ (discount factor)	—	0.99
Learning rate	—	0.005
Exploration noise σ_{noise}	—	$0.15 \to 0.02$ (annealed)
Training episodes	—	300
Eval. episodes (last k)	50 (last 30)	50 (last 30)
Seeds	10	5
R_{crit}	—	0.15
R_{recov}	—	0.25
p_{shock} / δ_{shock}	—	0.05 / 0.15
SVO θ range	$[0, \pi/2]$	—
Byzantine fraction	0.0–0.3	0.0–0.3

G Hyperparameter Summary

Paper Checklist

1. **Claims**: All main claims (Moral Commitment Spectrum, Nash Trap, unconditional commitment necessity) are supported by experiments within the tested PGG environments. We do not claim generality beyond the evaluated settings. [Yes]

2. **Limitations**: Discussed in Section 5.5. Key limitations include full observability assumption, heuristic design of $g(\theta, R)$, and migration rate constraint. [Yes]

3. **Theory**: Proposition 1 states assumptions (group replacement dynamics, migration bound) and reports empirically measured values. Full protocol in Appendix B. [Yes]

4. **Experiments**:

   (a) All hyperparameters listed in Table 10 (Appendix G). [Yes]

   (b) Error bars: mean $\pm$ std across seeds reported where shown. [Yes]

   (c) Seeds: 5–10 per experiment, specified per table caption. [Yes]

5. **Code and Data**: Anonymized code repository will be provided as supplementary material. All experiments are reproducible with the provided scripts. [Yes]

6. **Broader Impact**: Discussed in Section 6. No direct negative societal impacts anticipated from this foundational research. [Yes]

7. **Safeguards**: Not applicable (no human subjects, no deployment). [N/A]

8. **Licenses**: Code released under MIT license. No third-party datasets used. [Yes]

9. **New Assets**: Custom PGG environments described in Section 3.3. Will be released with code. [Yes]

10. **Human Subjects**: Not applicable. [N/A]

11. **Crowdsourcing**: Not applicable. [N/A]

12. **IRB**: Not applicable. [N/A]

13. **LLM Usage**: LLMs were used solely for proofreading and LaTeX formatting assistance. All scientific content, experimental design, and analysis are the authors' original work. [Yes]